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Resolving competing evolutionary histories in joint ancestral state reconstruction

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Ancestral state reconstruction (ASR) is a foundational tool in comparative biology, offering insights into the evolutionary history of lineages. With each new evolutionary model, our ability to estimate ancestral states with increased biological realism has improved. However, the field has primarily relied on marginal reconstructions, which focus on individual nodes. This framework is analytically tractable and appropriate for node-specific hypotheses, but it is not designed to identify the most probable sequence of evolutionary events across a tree. We argue that for researchers interested in evolutionary trajectories, joint reconstructions provide a more effective way to characterize the full history of transitions. Traditionally, joint reconstruction algorithms focused only on the single most likely sequence, but here we use conditional probabilities derived from stochastic mapping to sample the distribution of plausible ancestral histories efficiently. Furthermore, we provide tools to quantify and summarize this joint uncertainty. Through simulations and an empirical case study, we demonstrate that joint reconstructions more effectively recover simulated trait histories than node-wise marginal estimates and that the uncertainty surrounding these histories can be biologically meaningful. We apply our methods to epidemic multidrug-resistant *Klebsiella pneumoniae* and find that the evolution of antibiotic resistance is not a single narrative but a series of competing histories. Each of these histories exhibits distinct phenotype–genotype transitions that node-wise approaches would struggle to identify, yet have critical implications for predicting and understanding resistance evolution.

phylogenetic comparative methods | comparative genomics | antibiotic resistance | ancestral state reconstruction | *Klebsiella pneumoniae*

Significance

Understanding the evolutionary trajectories of complex traits, such as antibiotic resistance, is a critical challenge. Traditional methods characterize individual ancestors in isolation, but they do not necessarily identify the most probable evolutionary pathway. We provide a framework that shifts the focus from isolated nodes to the reconstruction of entire sequences of events. When applied to the epidemic pathogen *Klebsiella pneumoniae*, our framework reveals that there is not a single path to antibiotic resistance, but several competing ones. By distinguishing these alternative histories, our work identifies genetic changes linked to resistance that previous methods would not have detected, enabling more precise and testable hypotheses about pathogen evolution.

Ancestral state reconstruction (ASR) is central to many, if not most, comparative studies in evolution. Typically based on observed state data ranging from genomic sequences to geographic areas to phenotypic descriptions (1–3), a fitted model is used to infer the unobserved states of ancestral lineages. This practice has become a valuable tool for comparative biologists because it provides context for the order of evolutionary events, often representing the only window into the unobserved history of life. ASR can also generate testable hypotheses by inferring ancestral states, which can subsequently be synthesized in the laboratory (e.g., 4). For example, reconstructed ancestral steroid receptors have been used to determine when specificity for different hormones evolved and to demonstrate that several functional properties were lost in descendant lineages (5). Even when reconstruction is not the primary goal, ASR approaches are an essential intermediate step of many other comparative analyses (2, 3, 6).

The increasing integration of phylogenetic perspectives in medical sciences has underscored the value of ASR (2, 7). Phylogenetic comparative approaches have been successfully applied to model cancer prevalence and susceptibility across vertebrates (8), to identify ancient enzymatic functions for comparison with modern metabolic disorders (9, 10), and to reconstruct ancestral states of virulence factors and antibiotic resistance genes (11–13). Reconstructing ancestral states has proven particularly powerful for understanding the evolutionary history of antibiotic resistance. By distinguishing between de novo evolution within hosts and the transmission of resistant strains, ancestral reconstructions have improved our understanding of how antibiotic resistance emerges and spreads (14–16). Despite its increasing application, little attention has been given to whether the methodological foundations of ASR align with the biological questions that researchers seek to answer.

For historical and computational reasons, the field has primarily relied on specifying ancestral states based on marginal probabilities (2, 3, 17). Although ASR initially relied on parsimony algorithms for its estimates (e.g., 18), it quickly expanded to include maximum likelihood (19) and Bayesian (20) methods. In likelihood-based methods, a parametric model uses an empirical Bayes framework to estimate the posterior probability of

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each character state at a specific node, integrating over all possible scenarios at all other nodes (21). Essentially, for a given node, you are asking the probability that a state was present, considering the cumulative probability of all historical scenarios (Fig. 1). However, while marginal probabilities are optimized for testing hypotheses about individual ancestors, they are not designed to identify the most probable sequence of evolutionary events across a lineage (22–24). In contrast, joint reconstruction methods aim to estimate the most probable configuration of states across all nodes simultaneously (Fig. 1). This approach identifies the character history that maximizes the probability of the observed data, providing a more coherent representation of evolutionary pathways (25). While marginal reconstructions provide uncertainty in the form of probabilities for a given state, algorithms for estimating the joint reconstruction have focused primarily on finding only the single best reconstruction (24). Consequently, researchers often face a choice between node-specific probabilities that may lack path-wise coherence or a single joint history that lacks a measure of uncertainty (26). What is needed is a method for estimating and representing the uncertainty around a joint estimation, as well as the tools for describing and utilizing that uncertainty in an empirical setting.

We present a framework that uses the conditional probability logic of stochastic character mapping (27, 28) to quantify joint uncertainty for phenotypic and genomic information. Additionally, we develop a suite of methods for reconstruction clustering that facilitate the characterization of "major axes of ancestral reconstruction variation." The phylogenetically aware clustering approach allows empiricists to examine a representative set of alternative character histories. Each of these representative character histories can then be used as testable alternative hypotheses in downstream analyses. To evaluate our methodology, we first compare the ability of joint and node-wise marginal reconstructions to recover simulated trait histories. Finally, we develop several auxiliary tools necessary to apply these algorithms in an empirical setting and apply our framework to study the evolution of resistance to last-resort antibiotics in carbapenem-resistant *Klebsiella pneumoniae*.

Results

Performance of Joint and Marginal Reconstruction for Complete Character Histories. Our simulation study was designed to evaluate the effectiveness of joint and marginal ASR in recovering the true generating history across various evolutionary scenarios (*Materials and Methods*). Because marginal ASR optimizes the probability of states at individual nodes while joint ASR optimizes the probability of the entire sequence of states, we expect that joint reconstructions are better suited for identifying the true character history. Nonetheless, to demonstrate the magnitude of the discrepancy, we perform McNemar's tests (29) across our simulation conditions. The overall test revealed significant

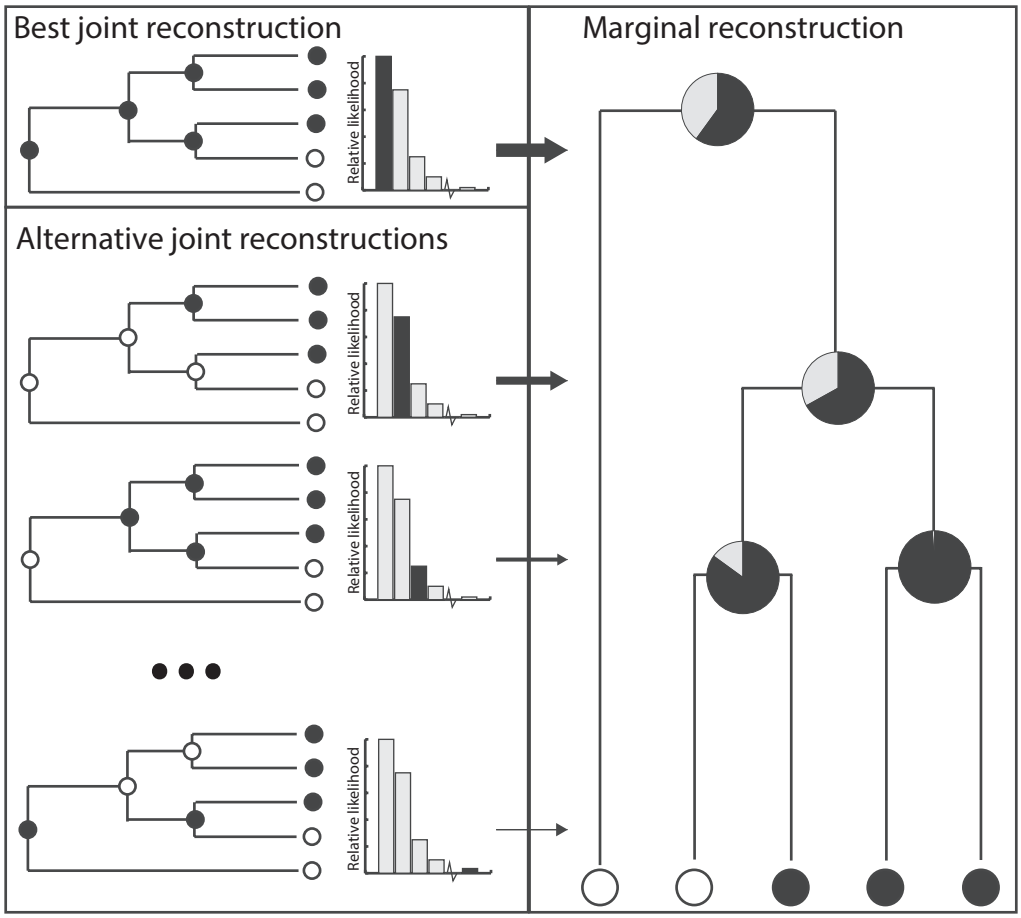


Fig. 1. The relationship between the joint reconstruction and marginal reconstruction. The marginal reconstruction can be considered an aggregation of all possible joint reconstructions. In doing so, the marginal approach integrates over all alternative histories to provide posterior probabilities for each node, whereas a joint reconstruction represents a specific character history. While standard algorithms typically only return the single best (maximum likelihood) joint reconstruction, there are often several other plausible reconstructions that are ignored. Even though these alternatives may have a lower relative likelihood, they can represent distinct evolutionary histories that differ from the single most likely scenario. These alternatives can change inferences concerning when and how many times a phenotype emerges, as well as influencing downstream analyses.

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differences between the methods (χ^2 = 88.804, P < 2.2e−16), with the joint reconstruction consistently identifying the true simulated history significantly more often than an assembly of the best marginal estimates. We compared this recovery rate across 100 simulated datasets for each parameter combination, finding that while both methods are sensitive to the underlying transition rates, the joint method more reliably identified the correct sequence of events.

At low transition rates (0.25), both methods performed similarly. In these scenarios, the historical signal is sufficiently strong that optimizing for individual nodes (marginal) or the entire path (joint) frequently leads to the same result (joint: 70.8%, marginal: 70.1%, McNemar's χ^2 = 1.89, P = 0.169). However, as the transition rate increased to medium levels (0.5), the advantage of path-wise joint optimization became more pronounced. As the complexity of the history increases, the conditional dependencies between nodes become more critical; in these cases, joint ASR maintained a significant edge in recovering the simulated configuration (two-state: 32.2% vs 30.5%, McNemar's χ^2 = 7.31, P = 0.007; three-state: 41.9% vs 40.9%, McNemar's χ^2 = 2.13, P = 0.144). At the highest transition rates (1.0), the amount of recoverable historical information is significantly diminished for any method (30). While both approaches performed poorly (recovery rates < 6%), joint reconstruction continued to identify the true configuration more reliably than the node-wise marginal assembly (two-state: 4.4% vs 3.6%, McNemar's χ^2 = 3.5, P = 0.061; three-state: 5.9% vs 5.0%, McNemar's χ^2 = 4.27, P = 0.039).

Joint Posterior Sampling Provides Robust Coverage of Evolutionary Pathways. To evaluate the reliability of our uncertainty quantification, we examined the frequency with which the true simulated history is captured within a finite distribution of M reconstructions. (recovery efficiency). By generating alternative configurations in proportion to their joint posterior probabilities, our framework reliably captured the true state configurations at low transition rates (95% of simulations for two-state and 92% of simulations for three-state recovered the generating character history). As expected, as transition rates increase and the recoverable historical information diminishes (30–32), the frequency with which the true history matches the single "best" point estimate declines. At intermediate rates, the joint reconstructions remain informative, and the true history was captured within the sampled joint distribution in 82% of simulations for two-state datasets and only 34% for three-state datasets. At the highest rates, nearly none of the true histories were captured in three-state datasets (2%), but 30% of the true histories were captured in two-state datasets.

In all cases, the results demonstrated the expected difference between joint posterior sampling and the naive independent sampling of marginal probabilities (Table 1). While both approaches can be used to generate a distribution of ancestral configurations, joint sampling is optimized to identify path-consistent histories. We observed that joint sampling identifies the true evolutionary trajectory with significantly higher efficiency than the marginal approach given the same maximum sampling effort. This efficiency arises because joint sampling focuses on configurations with high global likelihood, whereas independent node-wise sampling frequently generates "incoherent" histories that require a vastly larger number of samples to approximate the true posterior. These are sequences of states that may be likely at individual nodes but represent highly improbable transition sequences across the tree (Fig. 1).

Finally, we found that joint intervals are tightly centered around the true ancestral history and provide narrower, more precise credible sets than node-wise assemblies (Fig. 2). In contrast, independent marginal sampling tends to include a high proportion of improbable configurations that may not be path-consistent, resulting in a distribution that is less representative of the true evolutionary process. These findings indicate that joint uncertainty quantification provides a more efficient and biologically coherent estimation of ancestral state uncertainty, particularly when trait evolution is rapid or complex.

Reconstruction Uncertainty Reveals Distinct Patterns of Resistance to Last-Resort Antibiotics in *K. pneumoniae*. Alongside these simulations, we applied our methodology to study the evolution of antibiotic resistance. ASR has been extensively leveraged to identify genetic determinants and characterize the evolutionary trajectory of resistant lineages, largely due to detectable signatures of convergent evolution via spontaneous mutation or the acquisition of mobile genetic elements (11–13). Here, we applied our framework to characterize the evolution of resistance to last-line antibiotic, β-lactam/β-lactamase inhibitor (BL/BLI) combinations, in 412 carbapenem-resistant *K. pneumoniae* sequence type 258 isolates collected across a long-term acute care hospital network from 2014 to 2015 (33). Equipped with a preliminary understanding of the genetic basis (14), we evaluated whether reconstruction uncertainty would illuminate our understanding of the evolutionary dynamics of resistance to these promising antibiotics.

Joint reconstruction revealed the circulation of several resistant lineages, alongside a background of resistance acquired on terminal branches, suggestive of frequent de novo evolution of resistance (Fig. 3*A*). The uncertainty quantification method revealed many alternative evolutionary histories of BL/BLI resistance, with over nine thousand alternative reconstructions identified (Fig. 3*B*). As individually analyzing each of these possibilities is infeasible, we developed a clustering approach based on a distance metric for ancestral reconstructions (*Materials and Methods*). This clustering approach created a distance matrix based on the weighted differences between ancestral reconstructions, combining differences in character states and their phylogenetic distributions. We then applied t-distributed stochastic neighbor embedding (t-SNE) (34) and a density-based clustering algorithm (35) across 1,000 random seeds. Hierarchical clustering (36) was performed on a coassociation distance matrix to identify seven distinct clusters of reconstruction states (termed 'reconstruction axes') for BL/BLI resistance (Fig. 3*B*). We note that the exact number and composition of clusters can vary depending on the stochastic sampling of the joint uncertainty analysis, t-SNE clustering implementation, and hierarchical clustering algorithm choice. However, our chosen methodology yielded

Table 1. Comparison of joint posterior sampling versus independent marginal sampling across varying numbers of states (k = 2, 3) and transition rates (q)

		Joint sampling		Marginal sampling	
No. states	Transition rate	No. samples	Prop. Correct	No. samples	Prop. Correct
2	0.25	43.27	0.95	71.61	0.95
2	0.5	249.34	0.82	426.22	0.7
2	1.0	704.63	0.3	885.33	0.12
3	0.25	153.91	0.92	257.44	0.85
3	0.5	638.31	0.34	815.68	0.24
3	1.0	973.77	0.02	990.95	0.0

No. samples indicates the average number of unique samples required to reach a representative distribution of histories. Prop. Correct represents the proportion of simulation replicates in which the true generating history was contained within the sampled distribution.

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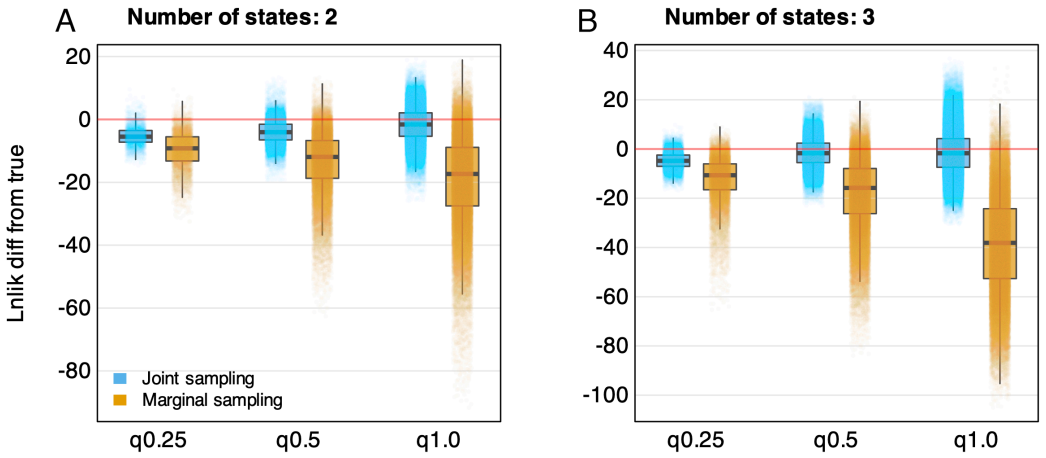


Fig. 2. Boxplots showing the distribution of log-likelihood differences between sampled character histories and the true generating history for (*A*) two-state and (*B*) three-state datasets across three transition rates (q = 0.25, 0.5, 1.0). The red line at 0 represents the likelihood of the true history; values below 0 indicate sampled histories with lower likelihoods than the truth. Joint sampling (blue) produces distributions centered around the true value as number of samples increases. Marginal sampling (orange) frequently generates histories with significantly lower likelihoods (large negative deviations), reflecting the "incoherent" combinations of states that arise when nodes are sampled independently. Jittered points represent individual simulation replicates; boxes indicate the interquartile range.

the most consistent results among several tested alternatives. Nonetheless, each reconstruction axis captured distinct reconstructions that described unique emergence events, altering the distribution of trait transitions and continuation events across the phylogenetic tree (Table 2). That is not to say there are no general patterns consistent across all reconstructions. Indeed, a visualization of important nodes with substantial variation across reconstruction axes revealed clade specific differences (Fig. 3 *A*–*C*). Reconstruction axes were inferred to have varying frequencies of resistance emergence, spread, and loss, predominantly due to variation in the ancestral node where resistance was acquired. Collectively, these findings support differential hypotheses regarding the evolutionary history of resistance, its associated fitness costs, and potential pathways for acquiring resistance to these antibiotics.

To test whether the different evolutionary histories found by our uncertainty framework influenced the appraisal of resistance-conferring genotypes, we used known resistance-associated genotypes as a form of truth testing. This analysis focused on several important genotypes implicated in BL/BLI resistance, including mutations to the *ompK36* porin, antibiotic efflux regulation (*ramR* efflux pump regulator and *ramA* efflux pump activator), target modification [penicillin-binding proteins (PBPs)], and antibiotic inactivation [*Klebsiella pneumoniae* carbapenemase (bla_{KPC})-containing plasmid, AA552, with elevated copy number].

We performed a pairwise comparison of phenotype and genotype reconstructions, quantifying the number of events in which the genotype and phenotype were gained along the phylogeny (Fig. 4). As expected, whether genotypic data aligned more closely with resistance emergence events depended on which reconstructions were used (Fig. 4*A*). Investigating the consistency of resistance acquisition across our genotypes revealed large differences in the consistency of synchronous transitions across reconstructions (Fig. 4*B*), suggesting that some reconstructions may more closely align with the dynamics of these resistance-associated genotypes. However, our pairwise integration of genotype and phenotype uncertainty also revealed several genotypes with minimal variation, and most reconstructions showed consistent, concordant predictions of trait evolution (Supplemental Materials). Finally, we investigated the utility of considering all genotypic reconstructions when explaining the evolution of resistance in this population. The proportion of phenotype gain events with at least one genotypic gain event varied across reconstruction axes. Notably, several reconstructions performed consistently better than the optimum joint reconstruction. Collectively, these findings suggest that evaluating uncertainty in phenotypic and genotypic reconstructions will improve our understanding of the evolutionary dynamics of a trait and will enable us to identify alternative hypotheses to consider for further testing.

Discussion

Reconstruction Uncertainty Impacts Genotype–Phenotype Associations. Phylogenetic approaches to genome-wide association studies often use synchronous transition events as evidence for genotype–phenotype relationships (37, 38). Our analysis of BL/BLI resistance in *K. pneumoniae* demonstrates that relying on a single "best" reconstruction may significantly impact these inferences. By applying our uncertainty quantification approach, we identified multiple plausible reconstruction axes, each of which infers a distinct evolutionary sequence for antibiotic resistance. These alternative pathways had meaningful consequences for investigating genotype–phenotype associations. Importantly, the frequency and location of synchronous phenotype–genotype gain events differed substantially between reconstructions, with some axes placing genotype transitions downstream of phenotype emergence. While this initially appears to be a biologically implausible scenario (suggesting the phenotype preceded its genetic cause), such patterns may reflect more complex evolutionary dynamics. For instance, downstream genetic variation could represent fitness compensation, fine-tuning of resistance mechanisms, or resolution of fitness tradeoffs following the primary resistance-conferring mutation (39–41). Crucially, these alternative reconstructions enable domain experts to evaluate competing evolutionary pathways using independent data, enhancing our understanding of resistance evolution.

Overall, these findings highlight a critical limitation of traditional approaches that select a single reconstruction. Focusing exclusively on the most likely configuration may lead researchers to overlook relevant genotypes simply due to the arbitrary selection of one evolutionary scenario among several plausible alternatives. For complex traits like antibiotic resistance, where multiple genetic mechanisms can produce similar phenotypes (42), this limitation is particularly problematic. By considering

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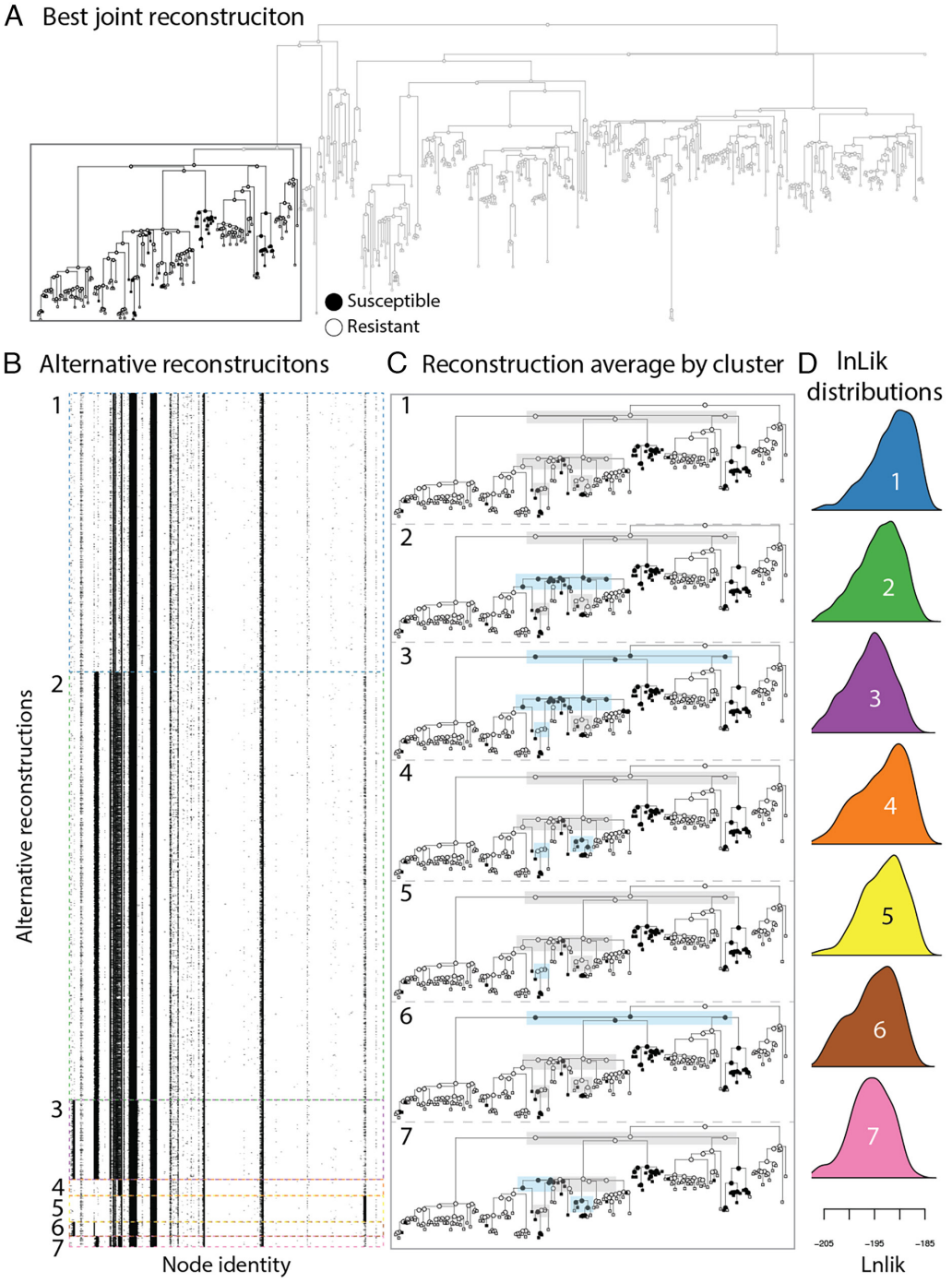


Fig. 3. Reconstruction uncertainty illuminates differential hypotheses regarding the evolutionary history of antibiotic resistance. (*A*) The maximum-likelihood phylogeny and best joint reconstruction of resistance to β-lactam/β-lactamase inhibitor among 412 carbapenem-resistant *Klebsiella pneumoniae* sequence type 258 isolates collected across a long-term acute care hospital network. Highlighted is the clade where most of the significant differences appear. (*B*) Alternative reconstructions where node states are flattened. Each row represents an alternative reconstruction sequence, and each column corresponds to a different node index. Reconstructions have been grouped into various clusters found by our phylogenetically aware clustering algorithm. (*C*) A representative reconstruction from each cluster, focusing on the most variable clade, as highlighted in panel *A*. Each reconstruction is chosen because it has the minimum average distance to all other reconstructions within the cluster. We have highlighted where key changes have occurred relative to the most likely joint reconstruction. A gray highlight indicates concordance with the most likely joint reconstruction and a blue highlight indicates where a susceptible reconstruction was estimated rather than resistance. (*D*) The likelihood distribution of the reconstructions within each cluster.

the distribution of plausible sequences when inferring associations, researchers can develop more robust models that better acknowledge the inherent uncertainty in ASR and capture the complexity of trait evolution.

An important avenue for future work is the development and application of correlated models that jointly infer the evolution of genotypes and phenotypes within a single framework. In our case, we modeled these components separately and then independently assessed relationships between events based on two distinct sets of ancestral reconstructions (one for genotypes and one for phenotypes). This approach requires considerable post hoc interpretation and multiplies uncertainty, since any inference

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Table 2. Summary statistics for phylogenetic reconstruction of antibiotic resistance events across different reconstruction axes

Reconstruction axis	Reconstructions	Singletons	Clusters	Cluster isolates	Gain events	Loss events	Continuations of trait
Reconstruction axis 1	3069	37.0 (33.0 to 41.0)	31.0 (25.0 to 36.0)	32.0 (26.0 to 38.0)	6.0 (4.0 to 9.0)	26.0 (20.0 to 32.0)	2.0 (2.0 to 5.0)
Reconstruction axis 2	4714	33.0 (26.0 to 40.0)	27.0 (19.0 to 33.0)	28.0 (20.0 to 35.0)	6.0 (3.0 to 10.0)	30.0 (23.0 to 38.0)	3.0 (2.0 to 8.5)
Reconstruction axis 3	874	30.0 (25.0 to 36.0)	26.0 (20.0 to 31.0)	27.0 (21.0 to 33.0)	4.0 (2.0 to 8.0)	31.0 (25.0 to 37.0)	2.0 (2.0 to 16.5)
Reconstruction axis 4	180	36.0 (32.0 to 39.0)	30.0 (24.0 to 33.0)	31.0 (25.0 to 35.0)	6.0 (5.0 to 9.0)	27.0 (23.0 to 33.0)	3.0 (2.0 to 4.0)
Reconstruction axis 5	288	36.0 (33.0 to 40.0)	29.0 (25.0 to 35.0)	30.0 (25.0 to 37.0)	7.0 (5.0 to 9.0)	28.0 (21.0 to 33.0)	2.0 (2.0 to 4.0)
Reconstruction axis 6	158	36.0 (33.0 to 39.0)	31.0 (25.0 to 35.0)	32.0 (26.0 to 35.0)	5.0 (4.0 to 8.0)	26.0 (23.0 to 32.0)	2.0 (2.0 to 4.0)
Reconstruction axis 7	117	34.0 (30.0 to 37.0)	27.0 (23.0 to 31.0)	28.0 (24.0 to 33.0)	7.0 (5.0 to 8.0)	30.0 (25.0 to 34.0)	3.0 (2.5 to 7.0)
Best joint reconstruction	1	37.0	31.0	32.0	6.0	26.0	2.5

Each row represents a different reconstruction axis, with the "Best joint reconstruction" showing the optimal joint reconstruction. Columns display: Singletons—number of phylogenetic singleton resistance events; Clusters—count of phylogenetic clusters of resistant isolates; Cluster isolates—number of isolates belonging to resistance clusters; Gain events—transitions from susceptible to resistant along edges of the phylogeny; Loss events—transitions from resistant to susceptible; Continuations—edges where the trait (resistance) persists. Values represent median estimates with ranges (minimum, maximum) across reconstruction iterations.

depends on reconciling multiple plausible genotype reconstructions with multiple plausible phenotype reconstructions. By contrast, models of correlated evolution have been shown to outperform independent models and offer a direct way to assess evolutionary relationships between traits (32, 43–46). For ASR, such joint models would yield a single set of predictions, greatly reducing the interpretive complexity. Ultimately, this integrated approach would not only strengthen genotype–phenotype association testing but also yield a more coherent picture of how molecular and phenotypic traits evolve together.

The Case for Joint Ancestral State Estimation With Uncertainty. ASR can provide crucial insights into character evolution and serves as a foundation for many comparative analyses. However, it is also a conceptual misnomer (23, 24, 32, 47). The use of the word "reconstruction" implies more certainty than is often warranted, and some authors prefer the term ancestral state estimation instead (47). Throughout this text, we have used the term "reconstruction" primarily for historical consistency, but ancestral states are estimates derived from the predictions of a fitted model. When viewed as estimates, it is clear that ancestral states are inherently uncertain because parameters cannot be perfectly known from a finite sample (26, 48, 49). This uncertainty is further compounded by the statistical inconsistency of ASR. Because information decays over time, the effective sample size for deep nodes remains small regardless of the number of tips, meaning states near the root may never be known with absolute certainty (30, 31, 50, 51). This leads to a fundamental question: What kind of ancestral state uncertainty is most important for biological inferences?

In this text, we have made the case for joint reconstructions as the more biologically appropriate method for characterizing trait trajectories. This is because the unit of joint reconstructions is entire evolutionary histories rather than single nodes, and this better aligns with evolution as a contingent process (52). While marginal ASR provides the correct uncertainty when testing a hypothesis about a specific ancestor, our simulations demonstrate that, as expected, path-wise joint optimization is significantly more effective at identifying the true trait trajectory. Traditionally, the field has relied on marginal methods because they are computationally efficient ways to summarize all possible histories (3, 53). However, the resulting node-wise probabilities do not directly represent the conditional dependencies present in a phylogeny. In empirical practice, researchers often assemble the most probable marginal states into a single tree to represent a history – a procedure that is not mathematically optimized for sequences and can result in biologically improbable transitions (28). A crucial limitation of this "naive assembly" is that it fails to capture the covariation of states across the tree, which is often the very basis of biological hypotheses.

This highlights the need not only to quantify uncertainty but also to do so in a way that preserves the structure of an entire evolutionary history. While joint reconstruction algorithms have existed for decades, they have traditionally focused on identifying only the single maximum-likelihood history, offering a false sense of certainty (24). By utilizing the conditional probability framework of stochastic character mapping (27, 28), we can move beyond point estimates to sample the full distribution of plausible joint histories. As our simulations demonstrate, the distribution of sampled joint histories is more likely to contain the true evolutionary path than a distribution generated by independent node-wise sampling. Because joint sampling accounts for the conditional dependencies between ancestors and descendants, it produces a set of histories that are globally probable and more centered around the true state configuration.

Of course, using the full uncertainty of joint reconstructions presents its own set of practical challenges. If the output of an analysis is not one history but thousands, how can a researcher visualize, interpret, and draw conclusions from this information? The sheer number of alternative histories can be impossible to interpret directly. This is why we developed these analytical tools, such as phylogenetically aware clustering and dimensionality reduction techniques. These methods allow researchers to move beyond an overwhelming set of possibilities to a manageable and interpretable set of distinct "reconstruction axes" or representative histories. By identifying the major

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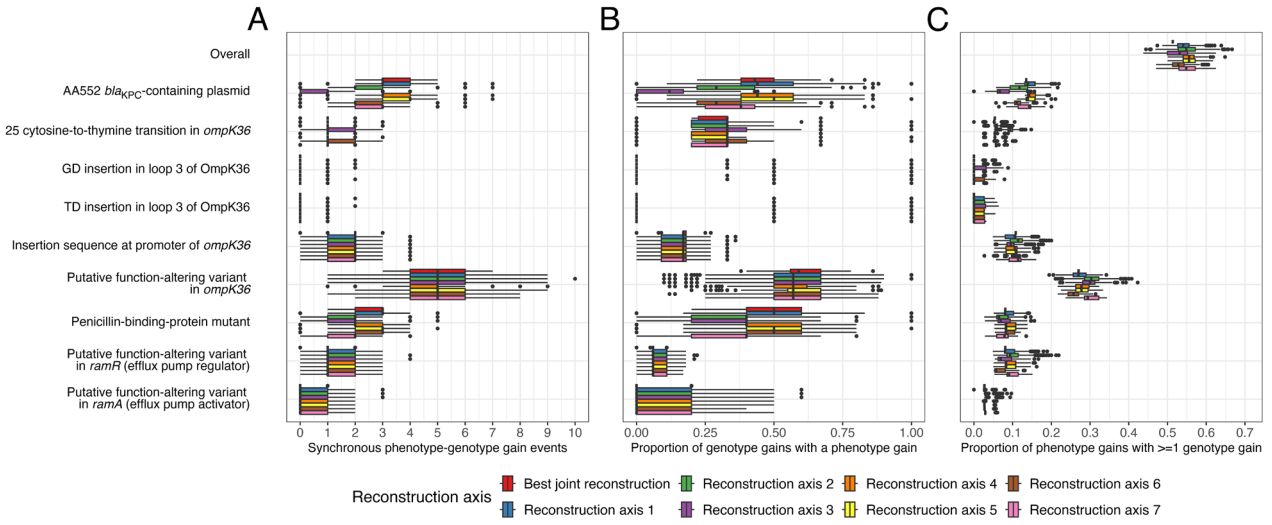


Fig. 4. Comparison of synchronous phenotype–genotype transition events across different evolutionary reconstructions. We compared reconstructions of our phenotype, β-lactam/β-lactamase inhibitor combination resistance, and carbapenem resistance-associated genotypes to detect synchronous transitions across the phylogeny, evidence of genotype–phenotype correlations. The figure displays distributions of (*A*) synchronous phenotype–genotype gain events (*B*) proportion of genotype gain events corresponding with a phenotype gain event and (*C*) the proportion of phenotype gain events in a reconstruction with ≥1 genotype gain event, considered individually and collectively. The inferred strength of any single phenotype–genotype association depends on the reconstruction axis chosen. Abbreviations: GD, Glycine-Aspartate; TD, Threonine-Aspartate.

patterns of variation among all plausible scenarios, these approaches distill immense complexity into a small number of alternative, testable hypotheses about how a trait may have evolved. This makes the theoretical advantages of joint uncertainty empirically tractable.

Finally, it is essential to acknowledge that these methodological considerations are embedded within a broader debate about the fundamental reliability of ASR (30, 54–57). Critics argue that ancestral state estimation is little more than informed guesswork, as its claims are difficult to confirm or refute without direct evidence from fossils. While it is undoubtedly important that ASR is approached with caution and confirmed with additional lines of evidence, we contend that the link between ancestral state estimates and the evolutionary model is not easily severed. If we regard ASR as inherently unreliable, we are implicitly suggesting that the models of trait evolution on which they are based are similarly detached from the biological reality they aim to represent. To reject ASR as "useless" is not just to critique one application, but to undermine the broader modeling framework that underpins phylogenetic comparative biology. That being said, there are technical challenges to ASR and the relationship between the model and the reconstruction is complex. For example, high transition rates provide significant information about the evolutionary process while simultaneously decreasing the information available for ancestral states (30). However, our point is focused more on the philosophical nature of this link than the technical one. If we believe these models represent the underlying stochastic processes of life, as we must to make inferences from them, then we should take seriously the predictions they make about character history. At the same time, if we accept that trusting model inferences means trusting their predictions, then technical insights like these help us identify when those predictions are most likely to be error-prone. A more productive approach is to acknowledge that ancestral reconstructions are model-based predictions subject to uncertainty and to evaluate them with the same rigor we apply to any other inference derived from evolutionary models.

Conclusion

Our results demonstrate that joint ASRs, combined with a framework for sampling and clustering joint uncertainty, provide more effective and biologically meaningful insights than traditional marginal approaches. While marginal methods are optimized to test hypotheses regarding individual ancestors, joint reconstructions treat entire evolutionary histories as the unit of inference. This perspective is consistent with the contingent nature of evolution and better aligns with the pathway-based questions that many biologists seek to answer. By applying the conditional probability logic of stochastic mapping to summarize alternative histories into major reconstruction axes, our framework enables uncertainty to be used to develop competing, testable hypotheses. Our empirical analyses of *K. pneumoniae* illustrate how this approach reveals distinct evolutionary narratives that are obscured by node-wise averages, with direct consequences for our understanding of genotype–phenotype associations. Ultimately, this framework enables biologists to treat ancestral state uncertainty as a collection of testable historical hypotheses grounded in the evolutionary model.

Materials and Methods

Background for Estimating the Probability of Ancestral States. The key distinction between a marginal and joint reconstruction is that the marginal marginalizes over all possible character histories to provide posterior probabilities for each node, while joint ASR identifies complete configurations of states across the tree. In both cases, estimating these probabilities requires specifying a model of evolution along the tree. The most common choice is a continuous-time Markov model, which defines transition probabilities along branches (43). This model is fit to the discrete character data and phylogeny and, once optimized, is used for downstream ASRs. Standard maximum-likelihood implementations of joint reconstruction (25) identify the single character history that maximizes the probability of the observed data but do not characterize the distribution of plausible alternatives. In contrast, efficient marginal reconstruction algorithms provide uncertainty at each node (17) but provide node-specific estimates that do not necessarily represent a globally probable sequence of events.

Consider an ASR problem with n nodes and k possible states. Let X_v represent the state of node v, where $v = 1, \ldots, n$. The state X_v can take on any value from

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the set $\{1, \ldots, k\}$. The marginal probability of node v being in state j is denoted as $P(X_v = j)$. This probability can be calculated by summing over all possible configurations of the other nodes:

$$P(X_v = j|Data) = \sum_{x_1=1}^{k} \cdots \sum_{x_{i-1}=1}^{k} \sum_{x_{i+1}=1}^{k} \cdots \sum_{x_n=1}^{k} P(X_1 = x_1, X_2 = x_2, \cdots, X_v = j, \cdots, X_n = x_n|Data),$$

where each x ranges over the possible states $\{1, \ldots, k\}$. In other words, this equation says that to find the marginal probability of node v being in state j, we consider all possible combinations of states for the other nodes and sum the posterior probabilities of each of those complete configurations. In contrast, the joint probability considers the probability of a specific configuration of states across all n nodes simultaneously. Let $x = x_1, \ldots, x_n$ represent a configuration specifying the state of each node. Following the logic of stochastic character mapping (27, 28), we can sample these configurations from their joint posterior distribution, $P(X = x|Data)$, to explore the uncertainty surrounding the evolutionary history.

Algorithm for Uncertainty Quantification. To characterize the uncertainty of joint reconstructions, we use the two-pass sampling framework established by stochastic character mapping (27, 28). This approach ensures that ancestral state configurations are sampled from their joint posterior distribution, $P(X|Data)$, rather than treating nodes as independent marginal estimates.

The first pass proceeds in a postorder fashion (from tips to root) following Felsenstein's pruning algorithm (58). For each node v, we calculate a vector of partial likelihoods, L_v, where $L_v(i)$ represents the probability of the observed data at the tips descending from node v, given that node v is in state i. For a tip, $L_v(i)$ is 1 if the observed state is i and 0 otherwise. For an internal node v with children y and z, the partial likelihood for state i is calculated as

$$L_v(i) = \left(\sum_{j=1}^{k} P_{ij}(t_y) L_y(j) \right) \left(\sum_{j=1}^{k} P_{ij}(t_z) L_z(j) \right),$$

where $P_{ij}(t)$ is the transition probability from state i to j over the branch length t.

The second pass proceeds in a preorder fashion (from root to tips) to sample a specific character history. First, the root state is sampled from its posterior distribution, which is proportional to the product of the root partial likelihoods and the root prior. Once the state of a parent node ($X_{anc} = i$) is determined, the state of its child node $v(X_v = j)$ is sampled from a conditional distribution. The probability of sampling state j for node x, given the parent is in state i, is

$$P(X_v = j|X_{anc} = i, Data) = \frac{P_{ij}(t_v) L_v(j)}{\sum_{m=1}^{k} P_{im}(t_v) L_v(m)}.$$

This preorder sampling is repeated down the tree until a state has been assigned to every internal node. By utilizing the partial likelihoods (L_v) calculated in the postorder pass, this procedure ensures that each node is sampled conditioned on both its parent's state and the observed data at the tips. Following this logic, we generate a collection of plausible joint reconstructions where each complete history is sampled in proportion to its joint posterior probability. Note that while these formulas are presented for bifurcating trees, they can be extended to account for polytomies by including additional child-term products in the partial likelihood calculation. This entire two-pass procedure is mathematically equivalent to the state-assignment phase of stochastic character mapping (27, 28).

Analyzing Patterns in Alternative Reconstructions. Because the number of potential character histories scales exponentially with tree size and state space, it is infeasible to evaluate every possible configuration. While individual visualization of reconstructions is sufficient when uncertainty is low, scenarios with substantial joint uncertainty require a method to identify and summarize the primary historical narratives present in the sample. To address this, we developed a phylogenetically aware clustering approach to identify distinct patterns among a sample of M alternative joint reconstructions while preserving biologically relevant variation. The method employs a three-stage analysis pipeline: distance calculation, dimensionality reduction, and hierarchical density-based clustering.

First, we quantified the dissimilarity between all M sampled histories. Each reconstruction, $R_k(fork = 1, \cdots, M)$, is a vector of character states $(X_{k,1}, X_{k,2}, \cdots, X_{k,N_{internal}})$ assigned to the $N_{internal}$ internal nodes of the phylogeny. Pairwise distances, $d(R_i, R_j)$, were computed using a custom phylogenetically informed metric (implemented as *corHMM::phylo_aware_dist_local*). For any two reconstructions R_i and R_j, let $V_{diff} = \{ v | X_{i,v} \neq X_{j,v} \}$ be the set of internal nodes where their assigned states differ. If the number of differing nodes, $|V_{diff}|$, is less than or equal to one, $d(R_i, R_j) = 0$. Otherwise, the distance is defined as

$$d(R_i, R_j) = \sum_{\{v_a, v_b\}} \exp\left(\frac{1}{d_P(v_a, v_b)} \right),$$

where the sum is over all unique pairs of distinct nodes $\{v_a, v_b\}$ within V_{diff}, and $d_P(v_a, v_b)$ is the patristic distance between nodes v_a and v_b on the phylogeny (calculated using *ape::dist.nodes* on a tree with all branch lengths set to 1). This metric emphasizes sets of co-occurring state differences at phylogenetically proximate nodes, assigning higher dissimilarity scores when multiplied differences are concentrated in specific areas of the phylogeny. This will yield D, an $M \times M$ symmetric distance matrix.

Second, the distance matrix, D, is used as input for t-distributed Stochastic Neighbor Embedding (t-SNE) to reduce dimensionality (34, 59). Default t-SNE parameters were used (via *Rtsne::Rtsne*), except to enable the custom distance matrix input (*is_distance = TRUE*). This transformation projects the high-dimensional space of reconstructions onto a two-dimensional manifold, attempting to preserve the local structure (i.e., similarities between reconstructions).

Next, Density-Based Spatial Clustering of Applications with Noise [DBSCAN; (35, 60)] was applied to the two-dimensional t-SNE coordinates to identify clusters of similar histories. The DBSCAN parameters were determined following established guidelines (35). The minimum number of points required to form a dense region (*minPts*) was set to 40, representing roughly 0.5% of the total number of reconstructions. The neighborhood radius (eps) was subsequently determined by examining a k-nearest neighbor distance plot (*dbscan::kNNdistplot*) for $k = minPts - 1$. The value of eps was chosen at the "knee" of this plot, which indicates a suitable threshold to distinguish informative points from noise. Due to the stochastic nature of the t-SNE algorithm, we repeated this process across 1,000 random seeds.

Next, a coassociation distance matrix was calculated to collate information of reconstruction clustering across each iteration. To generate the coassociation matrix, we calculated the probability of each reconstruction pair being found in the same cluster assignment. Hierarchical clustering was performed on a derived coassociation distance matrix using Ward's minimum variance method, as implemented in *stats::hclust* (*method = "ward.D2"*; 36). Final reconstruction clustering was determined after inspection of the hierarchical clustering dendrogram and clustering stability across multiple cut sizes.

To identify diagnostic nodes distinguishing between clusters, we developed a node importance metric for each node and each cluster (implemented as *corHMM::node_importance_by_cluster*): $I = (P_w)^2 * (1 - P_b)^2$, where I represents node importance, P_w is the proportion of matching ancestral states within a cluster, and P_b is the proportion of matches between clusters, calculated using Hamming distances from the modal within-cluster reconstruction. Nodes exceeding an importance threshold of 0.5 were considered diagnostic for their clusters. The *corHMM::plot_clustered_phylo* function visualizes these alternative reconstructions by displaying the proportion of states at each node within a cluster. The clustering process resolves the sample into distinct "global" narratives. Consequently, the pie charts within each cluster represent residual within-cluster variation rather than an average across all possible histories. This allows researchers to examine specific, path-consistent evolutionary scenarios that might be conflated or "blurred" when viewed through a single marginal summary.

Simulation Protocol for ASR Evaluation. We conducted a series of simulations to evaluate the recovery of complete character histories using joint and marginal ASR methods. Trees were simulated using the geiger R package (61) with a fixed number of taxa (ntaxa = 101 and nnode = 100) and a tree height of 1. Character data were simulated using the *corHMM* R package (62) under a continuous-time markov model with equal transition rates between states (ER model). To investigate the impact of transition rate and number of states on character history recovery, we varied the rate parameter (0.25, 0.5, 1.0) and the number of states (k = 2,3). These scenarios were chosen because transition rates are known to significantly influence

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the amount of historical information available for reconstruction (30, 32, 63, 64). For each combination of parameters, we simulated 1,000 replicate datasets.

For each simulated dataset, we fit an equal-rates (ER) model matching the generating process. Ancestral states were reconstructed using the *corHMM::ancRECON* function with parameters estimated from the data. To evaluate how these methods are typically used to interpret trait evolution in empirical studies, we defined the "marginal history" by selecting the state with the highest posterior probability at each node independently. We emphasize that while this node-wise assembly is a common practice among biologists, it is not mathematically optimized to identify a globally probable character history (Fig. 1).

To compare the history-recovery of joint and marginal ASR, we first determine the true ancestral state configuration for each simulated dataset. We then evaluated whether the joint and marginal reconstructions correctly identified this configuration. We summarized the results using a confusion matrix reporting the proportion of simulations in which 1) both joint and marginal methods recovered the true simulated history, 2) both failed to recover the true history, 3) marginal recovered the history while joint did not, and 4) joint recovered the history while marginal did not. Finally, to assess the sampling recovery efficiency of our uncertainty quantification, we examined the frequency with which the true simulated history is captured within a finite distribution of M reconstructions. This provides a measure of sampling efficiency, indicating how effectively each method captures the true history given a fixed computational effort. Additionally, we calculate a likelihood distribution of possible joint reconstructions based on our uncertainty quantification methods and examine how often the exact likelihood of the true ancestral states was found within this distribution.

Ancestral State Analysis of β-Lactam/β-Lactamase Inhibitor Resistance in Carbapenem-Resistant *K. pneumoniae*. We applied our uncertainty methodology to evaluate antibiotic resistance patterns in *K. pneumoniae*. This analysis focused on resistance to recently introduced β-lactam/β-lactamase inhibitor (BL/BLI) antibiotics within a densely sampled population of 412 carbapenem-resistant *K. pneumoniae* isolates belonging to the epidemic lineage, sequence type 258 (33). Briefly, carbapenem-resistant *K. pneumoniae* identified from clinical isolates among long-term acute care patients were subjected to short-read whole-genome sequencing on an Illumina HiSeq 2500 instrument. Use of deidentified genomes and metadata was approved by the Institutional Review Board at the University of Michigan. A Gubbins-recombination filtered maximum-likelihood phylogeny was reconstructed using IQ-TREE version 2.4.0 using the GTR+F+ASC+R2 substitution model. IQ-TREE was performed using default parameters, other than the requirement of 500 unsuccessful iterations to halt reconstruction. A total of 5,000 bootstraps were performed. More information on the sample collection can be found elsewhere (33).

Phenotypic resistance was measured for two BL/BLI agents, meropenem–vaborbactam and imipenem–relebactam, using the SENSITITRE broth microdilution (Thermo Fisher; 55). Resistance to BL/BLI agents was determined as a composite of resistance to these agents, per 2021 Clinical and Laboratory Standard Institute interpretative criteria.

The evolutionary history of BL/BLI resistance was evaluated using ASR. First, joint reconstruction was performed using corHMM under an equal rates model, with root prior estimated using the "maddfitz" procedure which treats the root state frequencies as nuisance parameters that are integrated out of the likelihood calculation (65, 66). Next, uncertainty in joint reconstruction was evaluated using corHMM's *compute_joint_ci*() with a batch size of 100 and a max sampling of 10,000 using the "simmap" procedure.

To illustrate the influence of ASR classifications on evolutionary inferences, we leveraged the open-source R package, *phyloAMR* (15; https://github.com/kylegontjes/phyloAMR). To trace the ancestral state of each isolate, we applied a tree traversal algorithm, *asr_cluster_detection*(). Using ancestral states, the approach walks from the root to every tip, looking for evidence of trait continuation (e.g., susceptible to susceptible), gain (e.g., susceptible to resistance), or loss (e.g., resistance to susceptible). Next, the algorithm takes each trait-containing tip and walks upward on the tree to classify strains as either phylogenetic singletons or members of a phylogenetic cluster associated with the trait. Trait-containing tips with gain events inferred at the tip were classified as phylogenetic singletons if there was no history of ancestral resistance deeper in the tree. However, tips with a trait and a gain event at the tip and a trait reversion event at their parent node were eligible for classification as a cluster isolate. Tips were classified as members of a phylogenetic cluster if their first inferred ancestral gain event was shared with at least one additional isolate with a trait. Isolates with the trait that did not share a gain event with any other resistant isolate were classified as phylogenetic singletons. Lineages, where all isolates belonged to one patient, were not considered evidence of phylogenetic clustering and were classified as redundant singletons. The algorithm's two identified states, singletons and clusters, permit distinguishing between de novo emergence of a trait and the transmission of trait-containing lineages, respectively. This algorithm was leveraged to test the inferences of resistance dynamics, notably number of phylogenetic events, gain/loss events, and size of clusters, across the alternative reconstruction axes inference identified using the clusters from the uncertainty algorithm.

To contextualize antibacterial resistance, we leveraged a collection of known carbapenem resistance-associated genotypes and a dominant bla_{KPC}-containing plasmid previously described elsewhere (14). Individually identified single-nucleotide polymorphisms, insertion/deletions, and insertion sequences were simplified through binary classification (e.g., present/absent). Specifically, genetic variants were grouped by gene to enhance detection power for genotype–phenotype associations. For example, multiple nonsynonymous mutations in the *ompK36* porin were consolidated into a single binary trait, as individual mutations might be too rare for meaningful statistical analysis.

Finally, we investigated the temporal relationship between genotype and phenotype transitions using the joint uncertainty and tree traversal algorithms. Phylogenetic transitions in genotype and phenotype are often used in genome-wide association studies to identify genotype–phenotype correlations (67, 68). We hypothesized that ancestral state differences influence the robustness of identified genotype–phenotype associations. To test this hypothesis, we evaluated the frequency of synchronous gain events of BL/BLI resistance and our resistance-associated genotypes using *phyloAMR::synchronous_transitions*() algorithm. We also evaluated the consistency of genotype gain events with resistance acquisition and the proportion of phenotypic gain events where at least one genotype reconstruction inferred a genotype gain event, individually and collectively across all genotypes. All code for this empirical case study is hosted at https://github.com/kylegontjes/joint-uncertainty-ms.

Data, Materials, and Software Availability. R code, phylogenetic trees, resistance data; All code for this empirical case study data have been deposited in Data dryad (69) and Github (70).

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